Mouhamadou Lamine SOW

Computer Science Engineering Student — Machine Learning & Operations Research Internship

Final-year internship • 6 months • from March 2027

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Profile

Final-year engineering student at Sup Galilée, working at the intersection of **machine learning** and **operations research**: predict with ML models, then decide with MILP and metaheuristics. Applied at **Orange (Sonatel)** and on the international **ROADEF/EURO 2020 challenge** (a real RTE problem).

Professional Experience

Machine Learning / Data Science Intern May 2026 – Jul 2026
Orange (Sonatel) Dakar, Senegal

- Designed an XGBoost model predicting network risks 6 months ahead: **88 % fewer missed at-risk cases** than the existing business rule (PR-AUC 0.81 vs. 0.28).
- Aggregated risk by area over **4 horizons from 3 to 12 months** to **prioritize field interventions**; calibrated, explainable probabilities (SHAP).
- Built, with a teammate, an **8-year fiber network history** (2018–2026); engineered temporal and geographic features with leakage-free chronological evaluation.

Job-Shadowing Internship — Digital Technology and DevOps Jul 2024 – Sep 2024
Orange (Sonatel) Dakar, Senegal

- Wrote a literature review on **intelligent DevOps automation**; introduction to Angular, LoopBack and MongoDB.

Projects

ROADEF/EURO 2020 Challenge — RTE Maintenance Scheduling 2026

Tools: Python, Gurobi, MILP, simulated annealing, local search

- Reached solutions **within 0.52 % on average of the best known result** of this international challenge, on 12 test instances (vs. 2.83 % for the greedy baseline).
- Designed a greedy repair heuristic followed by simulated annealing: **100 % feasible solutions**, verified with the official checker.
- Built a Gurobi MILP model strengthened with valid inequalities: **proven optimal on 8 of 15 instances**.

Ring-Star — Transport Network Optimization Academic project

Tools: Python, PuLP, Gurobi, CPLEX, CBC

- Formulated a ring-star network design problem as a **MILP** (station selection, ring routing, user assignment), solved with Gurobi, CPLEX and CBC.
- Developed a heuristic and a local search: **19 % lower cost** than the greedy approach, in under one second.

Education

Engineering Degree in Computer Science — Master's level, expected 2027 Sep 2024 – Present
Sup Galilée — Université Sorbonne Paris Nord Villetaneuse, France

Key coursework: operations research, combinatorial optimization, algorithms, machine learning, deep learning.

Integrated Preparatory Program (CPI) — MPSI Sep 2022 – May 2024
ESIGELEC–ESMT Dakar, Senegal

Mathematics, physics and engineering sciences.

Skills

Operations Research	MILP, combinatorial optimization, metaheuristics (simulated annealing), local search
Solvers	Gurobi, CPLEX, CBC, PuLP
Machine Learning	scikit-learn, XGBoost, PyTorch; imbalanced classification, time-based validation, calibration, SHAP
Programming & Data	Python, SQL, Julia, C/C++; pandas, NumPy, SciPy
Tools	Git, Linux, pytest, Jupyter

Languages & Certification

French: native **English:** B2 — TOEIC 802

IBM AI Engineering Professional Certificate — IBM / Coursera, in progress (2026).